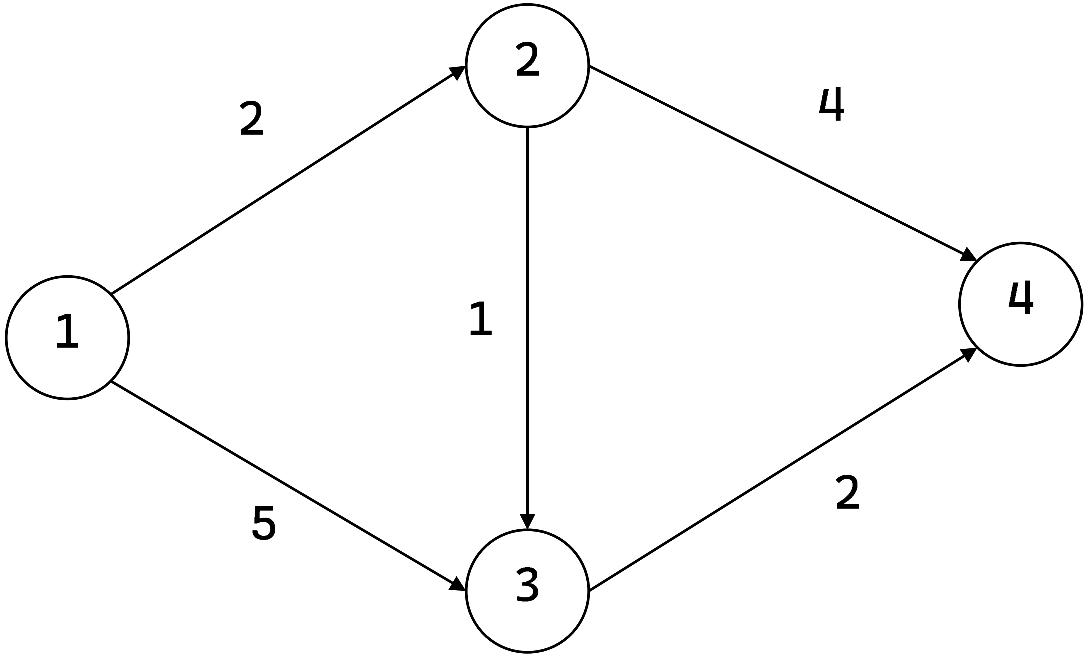
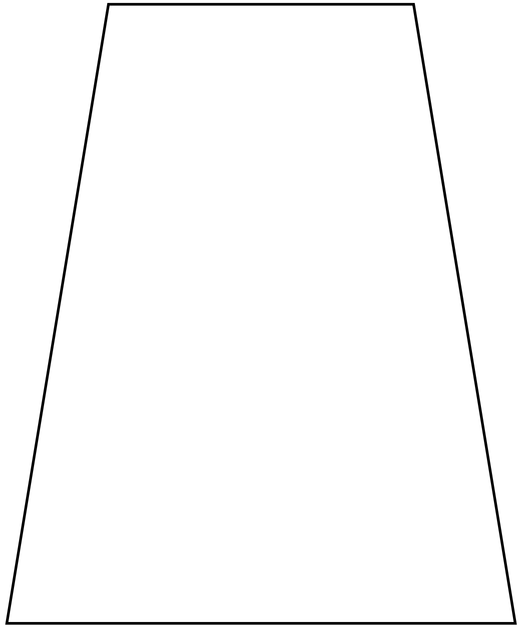
dis					
	index	1	2	3	4
	value	INF	INF	INF	INF



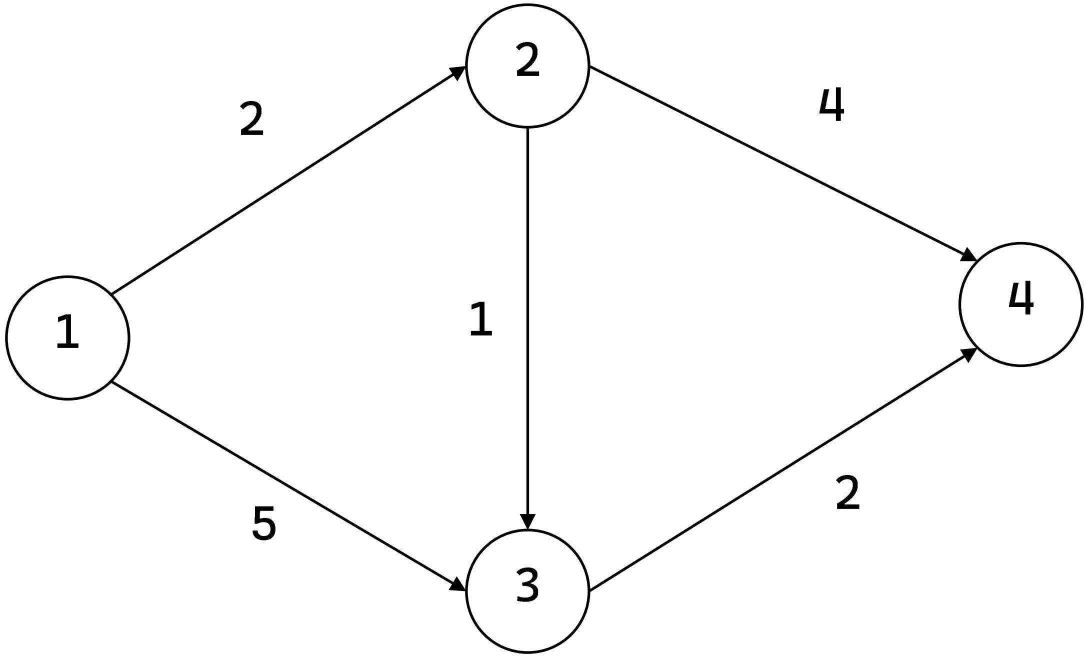
priority_queue
{與 1 的距離,點}



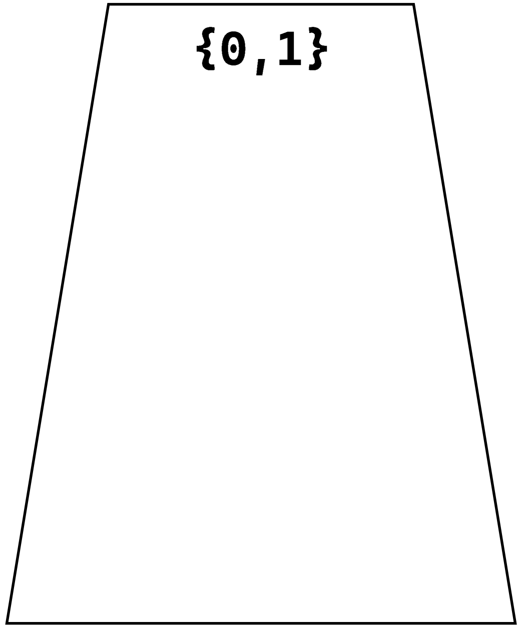
```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

初始化

dis					
	index	1	2	3	4
	value	0	INF	INF	INF



priority_queue
{與 1 的距離,點}

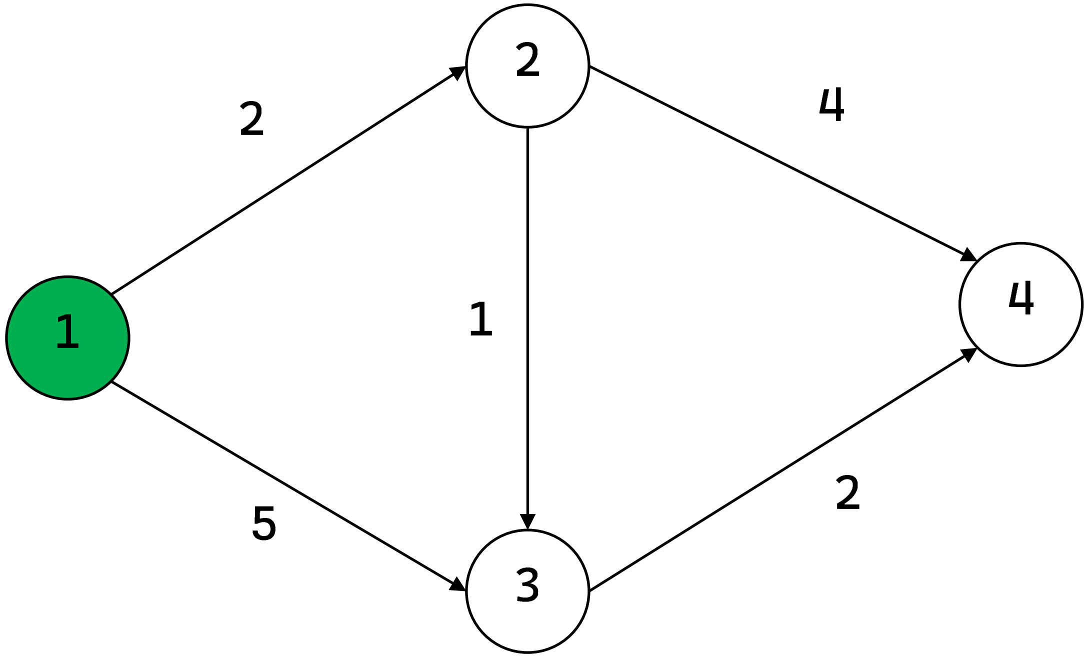



```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

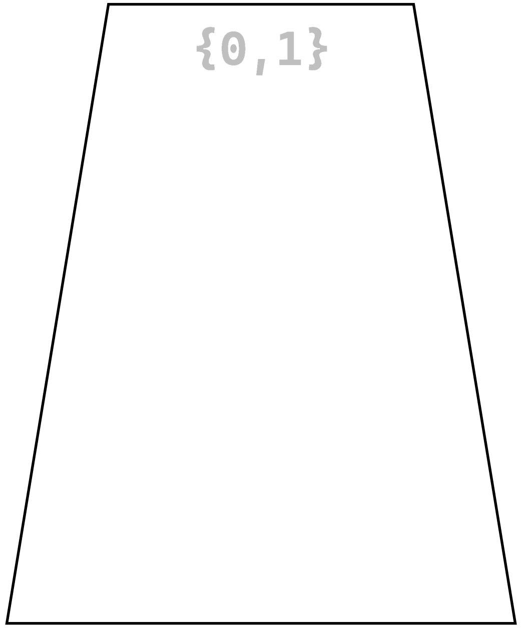
更新路徑：**dist[1] = 0**，並 **push {0,1}**

dis

index	1	2	3	4
value	0	INF	INF	INF



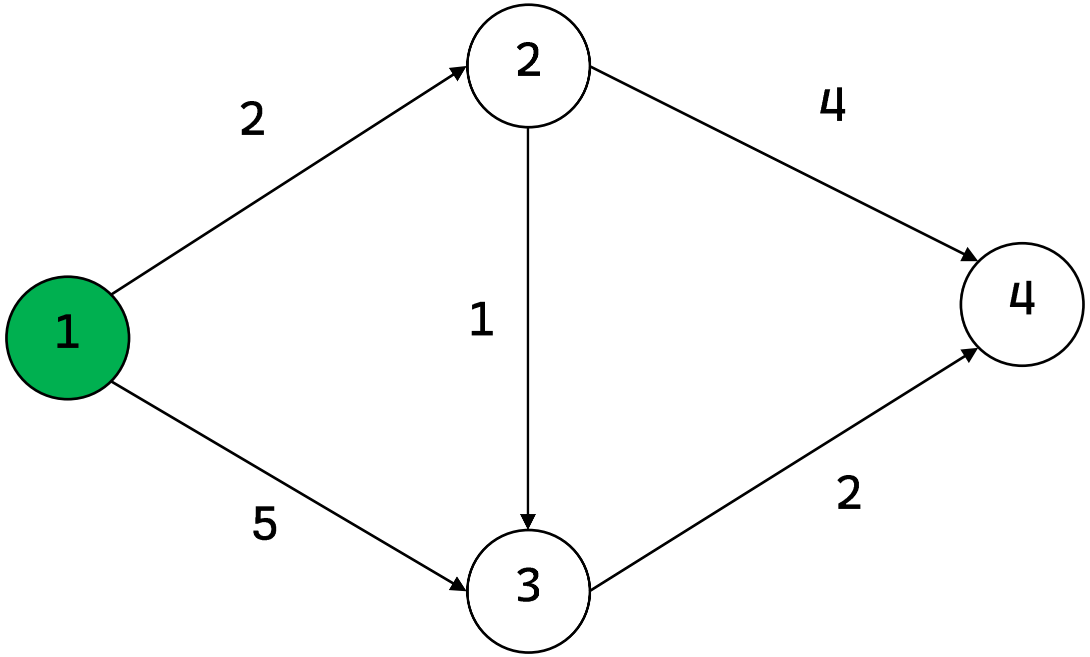
priority_queue
{與 1 的距離,點}



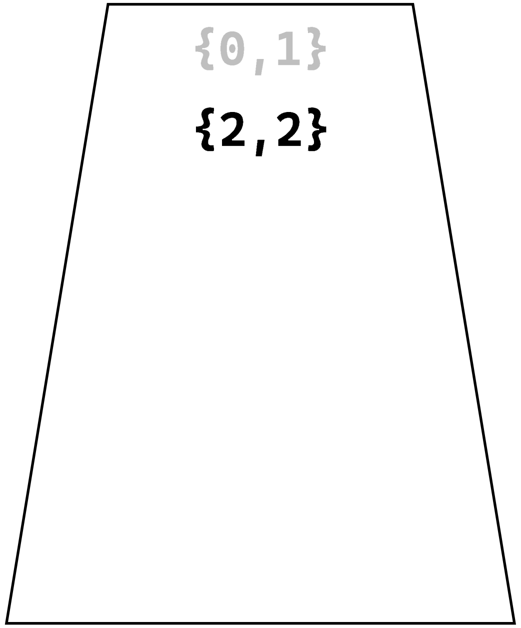

```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

取出最小的：**{0,1}**

dis				
index	1	2	3	4
value	0	2	INF	INF



priority_queue
{與 1 的距離,點}

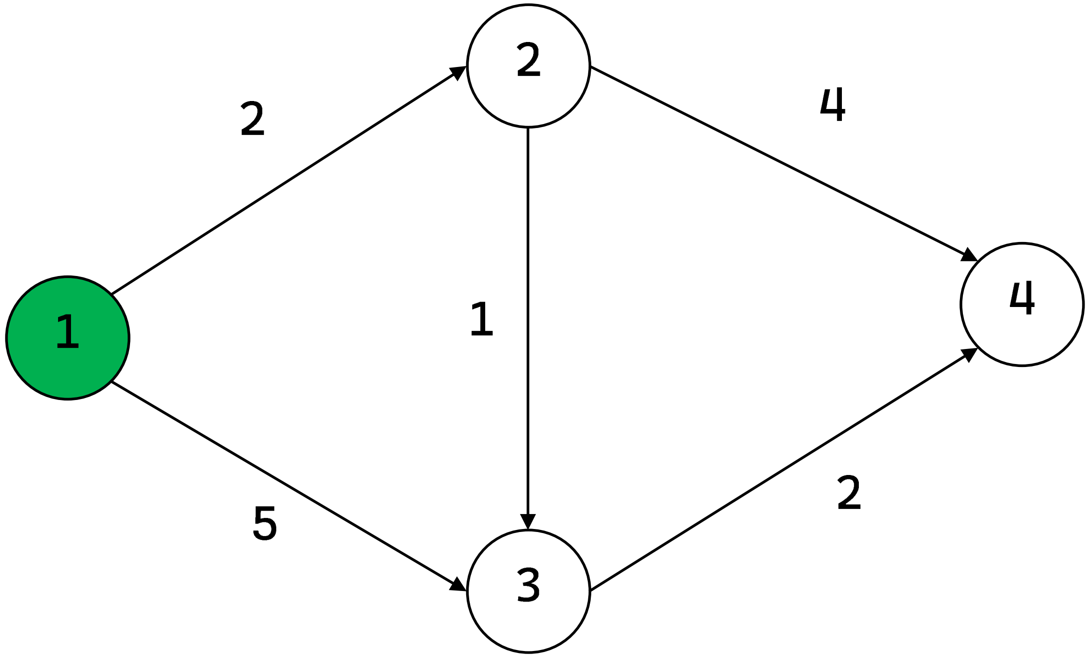



```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

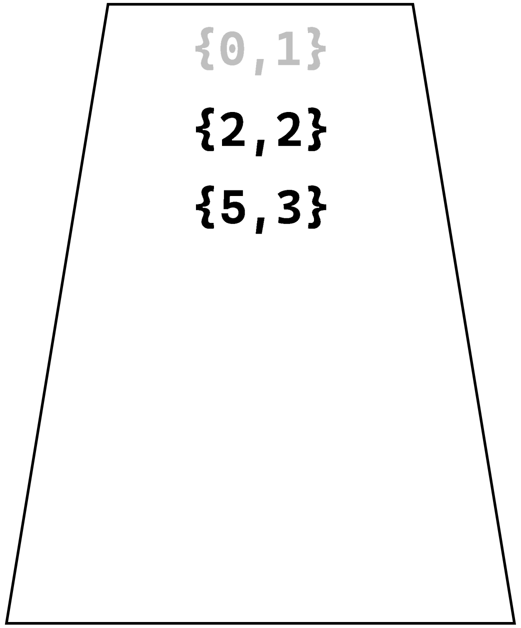
更新路徑：**dist[2] = 2，並 push {2,2}**

dis

index	1	2	3	4
value	0	2	5	INF



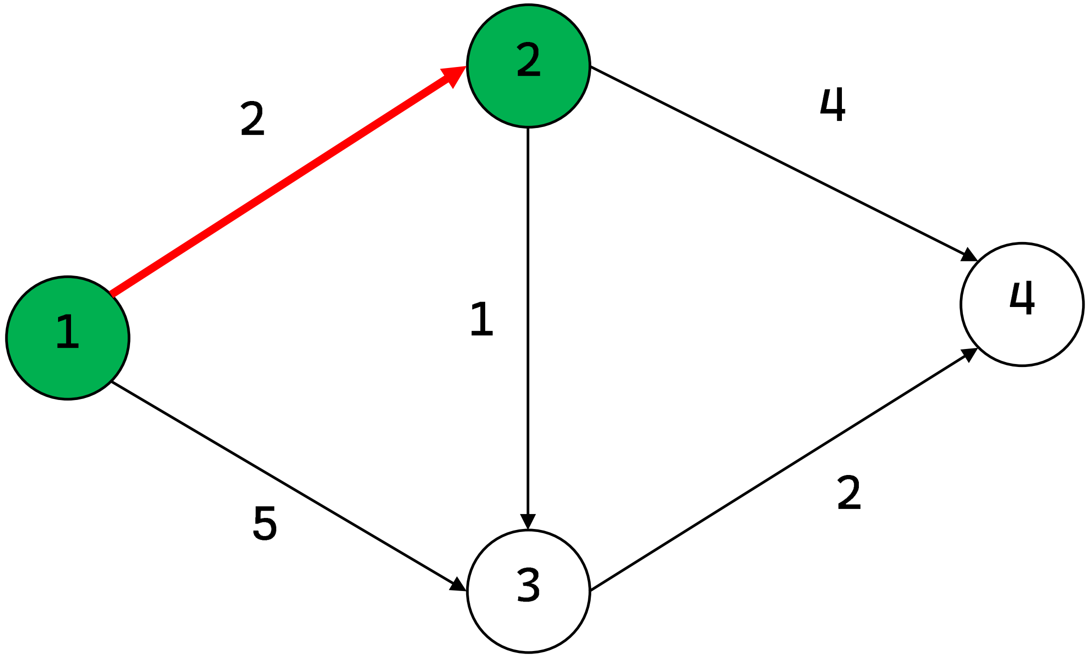
priority_queue
{與 1 的距離,點}



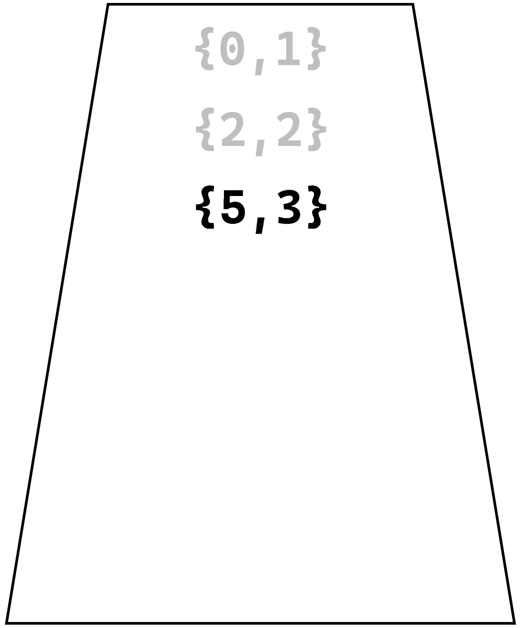

```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

更新路徑：**dist[3] = 5**，並 **push {5,3}**

dis					
	index	1	2	3	4
	value	0	2	5	INF



priority_queue
{與 1 的距離,點}

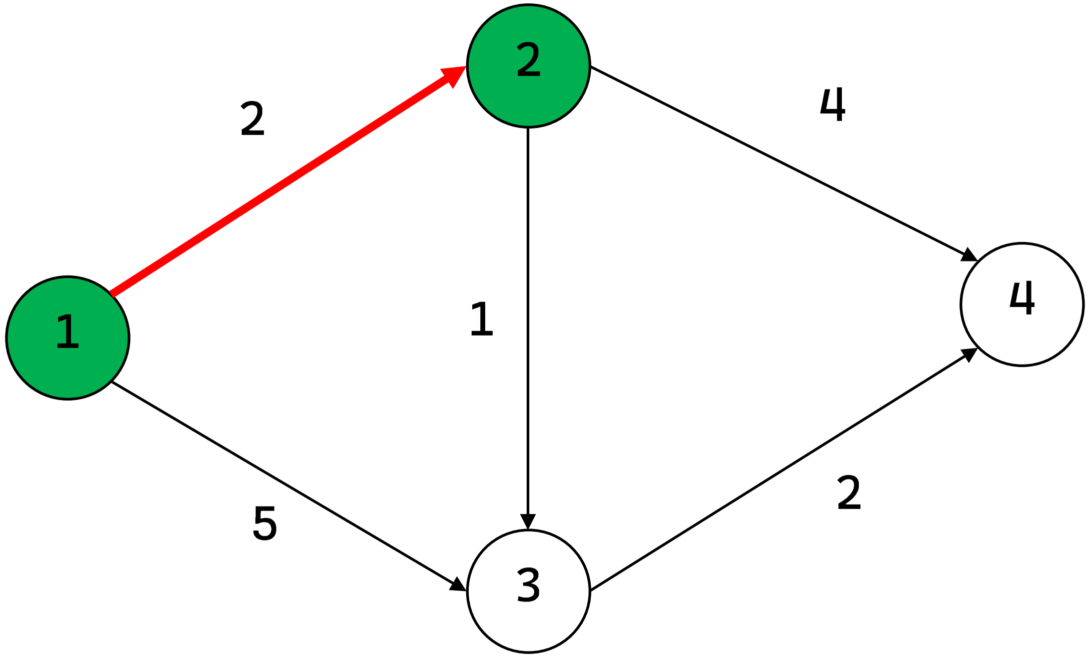



```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

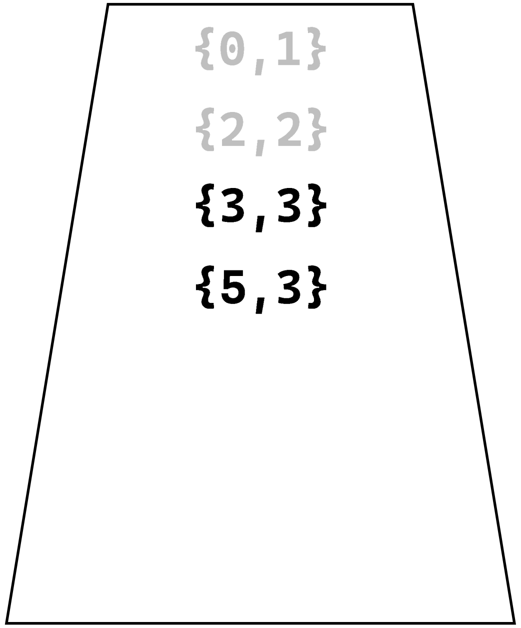
取出最小的：**{2,2}**

dis

index	1	2	3	4
value	0	2	3	INF



priority_queue
{與 1 的距離,點}

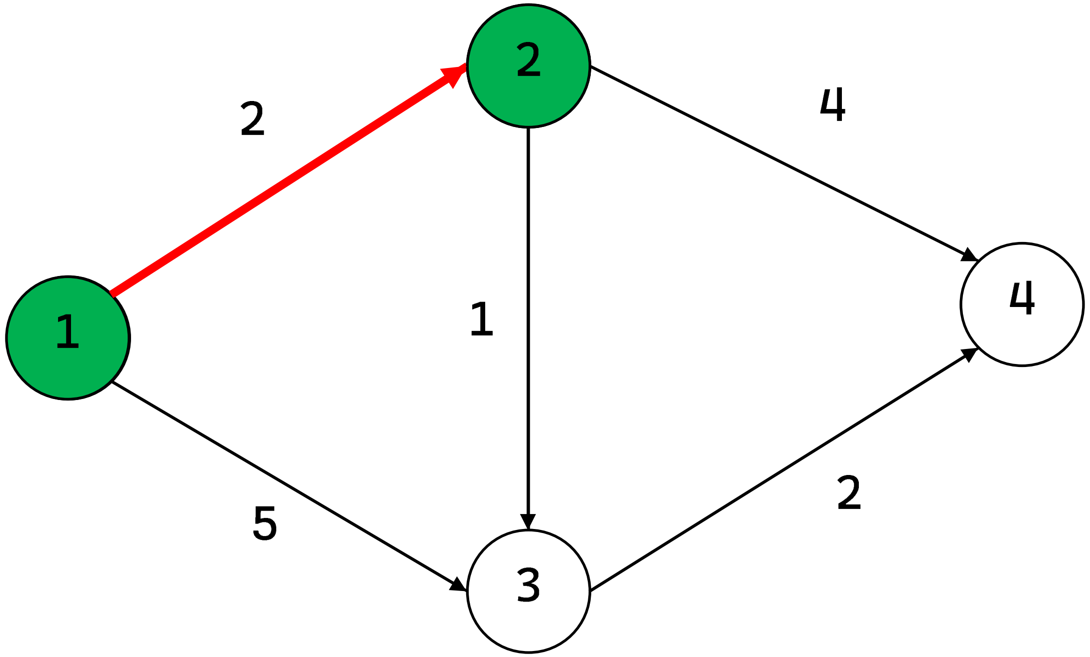



```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

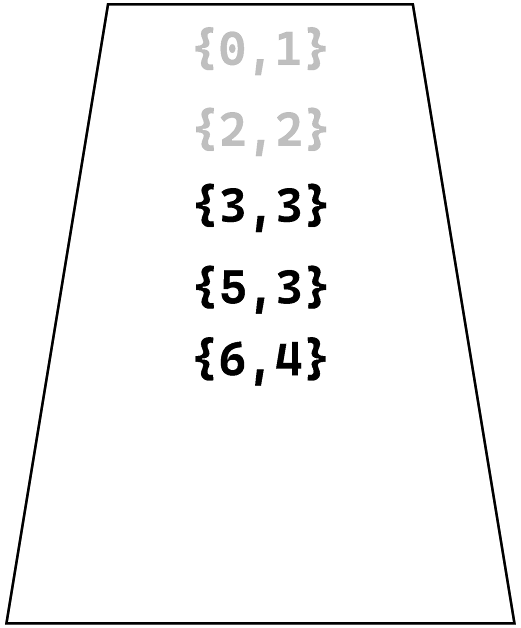
更新路徑： **dist[3] = 3** ，並 **push {3,3}**

dis

index	1	2	3	4
value	0	2	3	6



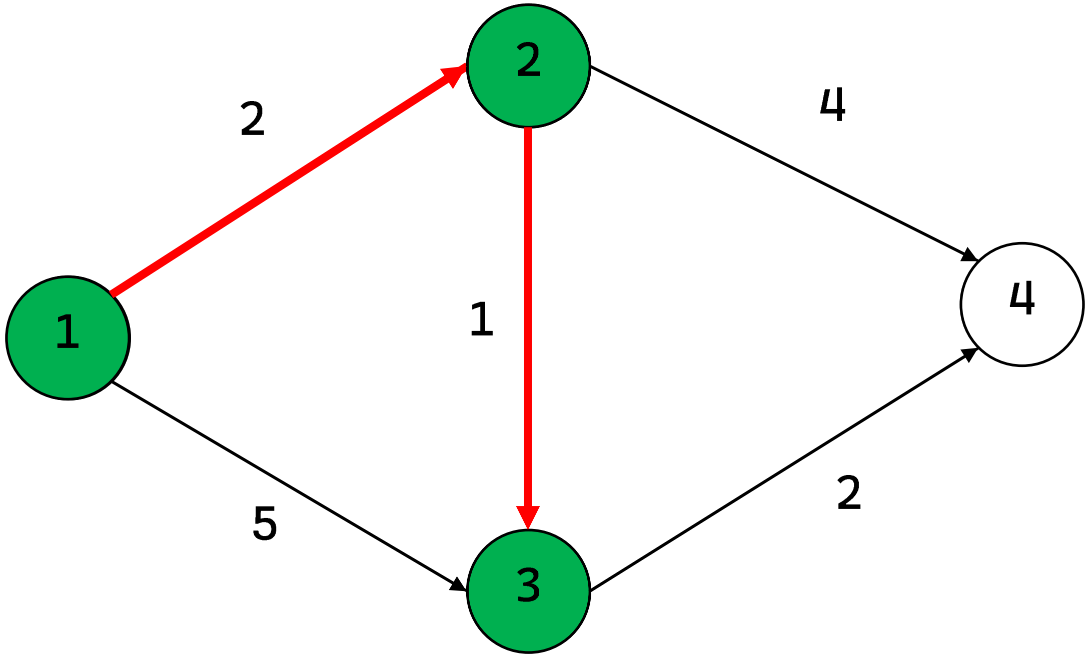
priority_queue
{與 1 的距離,點}



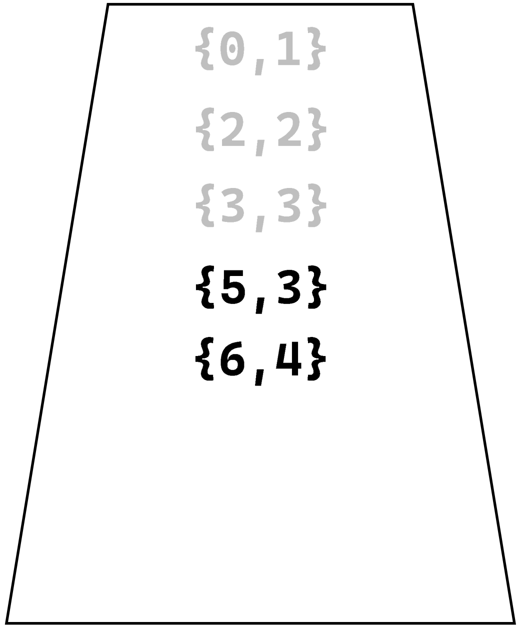

```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

更新路徑： **dist[4] = 6** ，並 **push {6,4}**

dis					
	index	1	2	3	4
	value	0	2	3	6



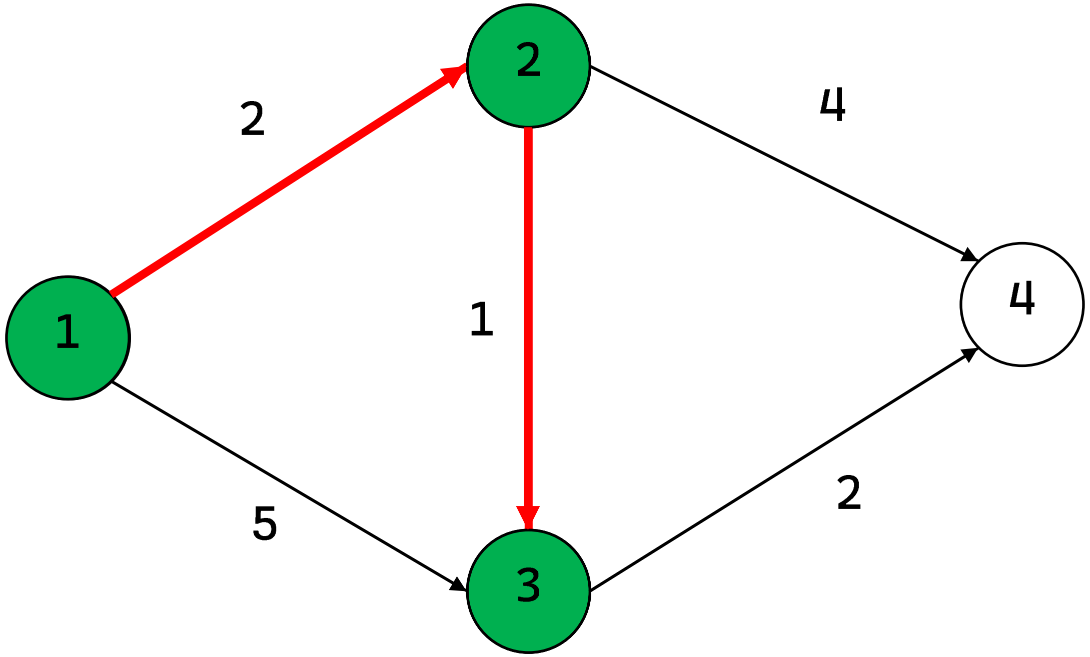
priority_queue
{與 1 的距離,點}



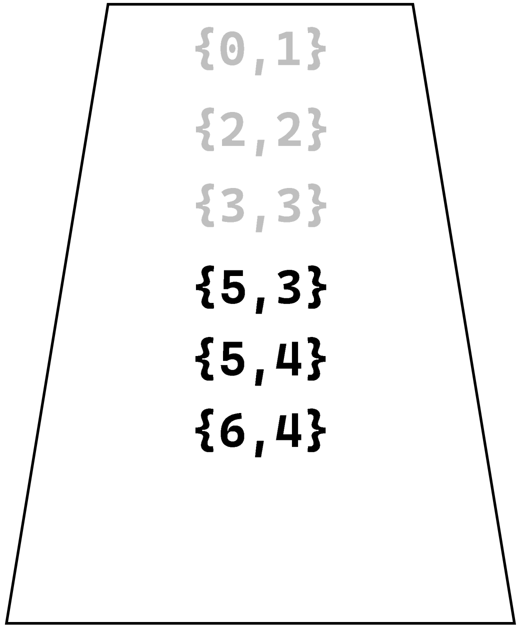

```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

取出最小的：**{3,3}**

dis					
	index	1	2	3	4
	value	0	2	3	5



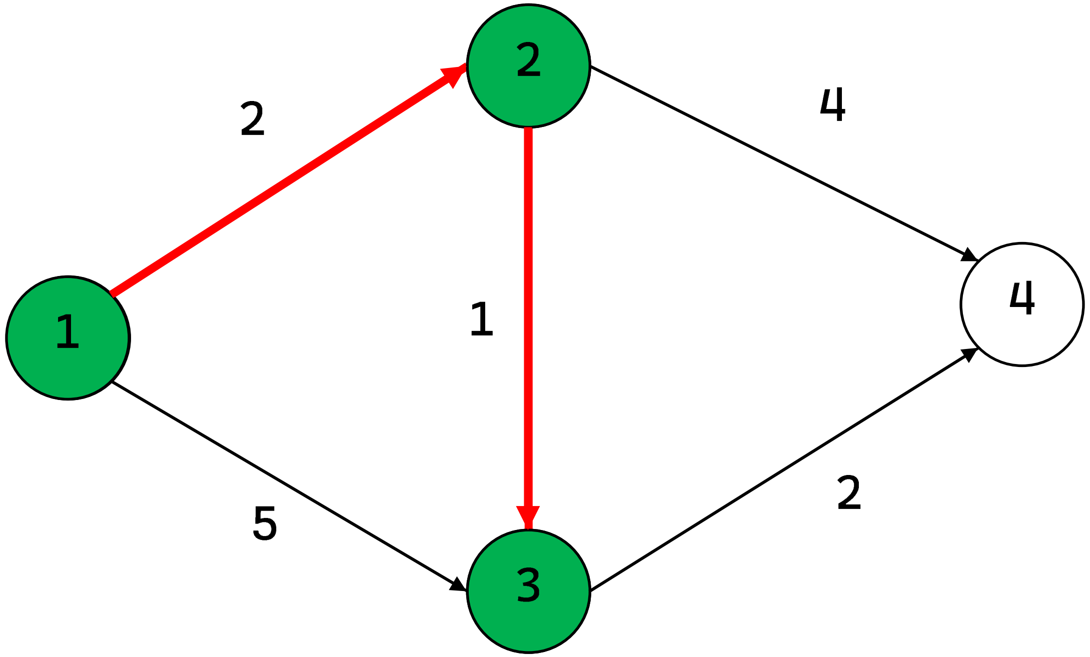
priority_queue
{與 1 的距離,點}



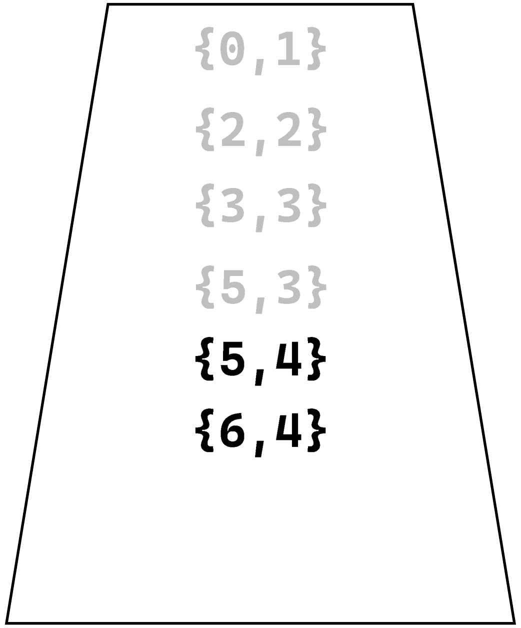

```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

更新路徑： **dist[4] = 5** ，並 **push {5,4}**

dis					
	index	1	2	3	4
	value	0	2	3	5



priority_queue
{與 1 的距離,點}

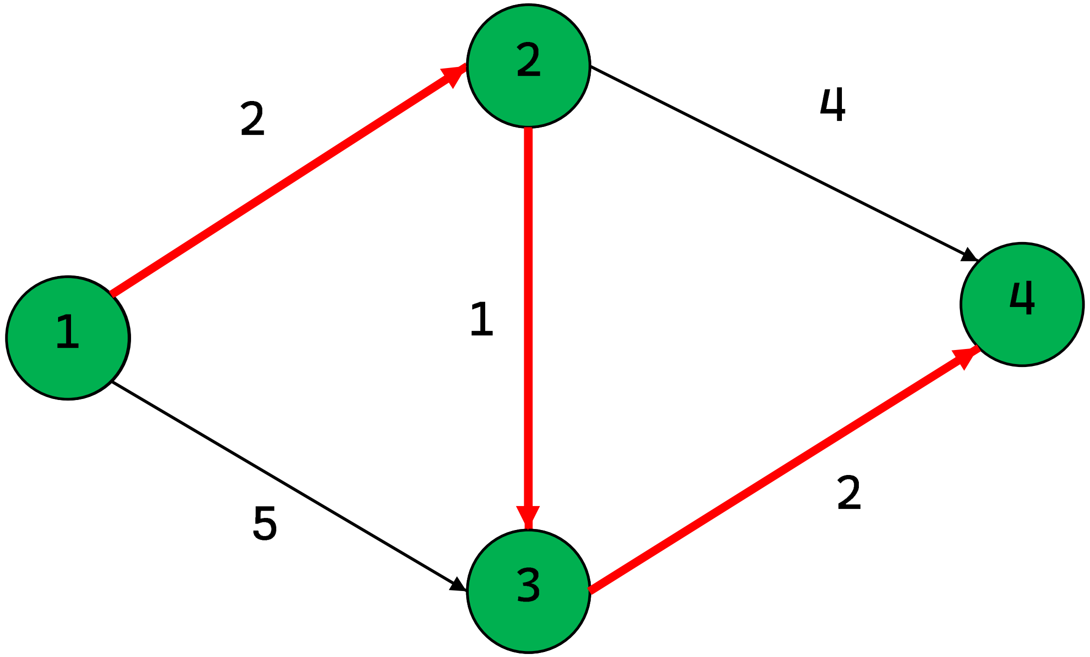



```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

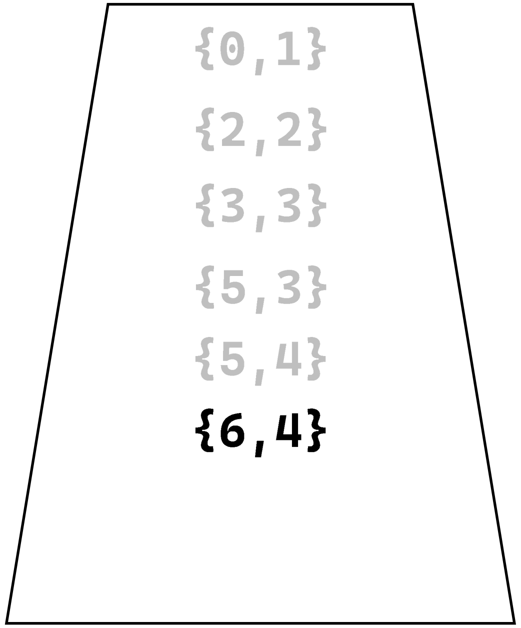
取出最小的：**{5,3}**，**5 ≠ (dis[3] = 3) → continue**

dis

index	1	2	3	4
value	0	2	3	5



priority_queue
{與 1 的距離,點}

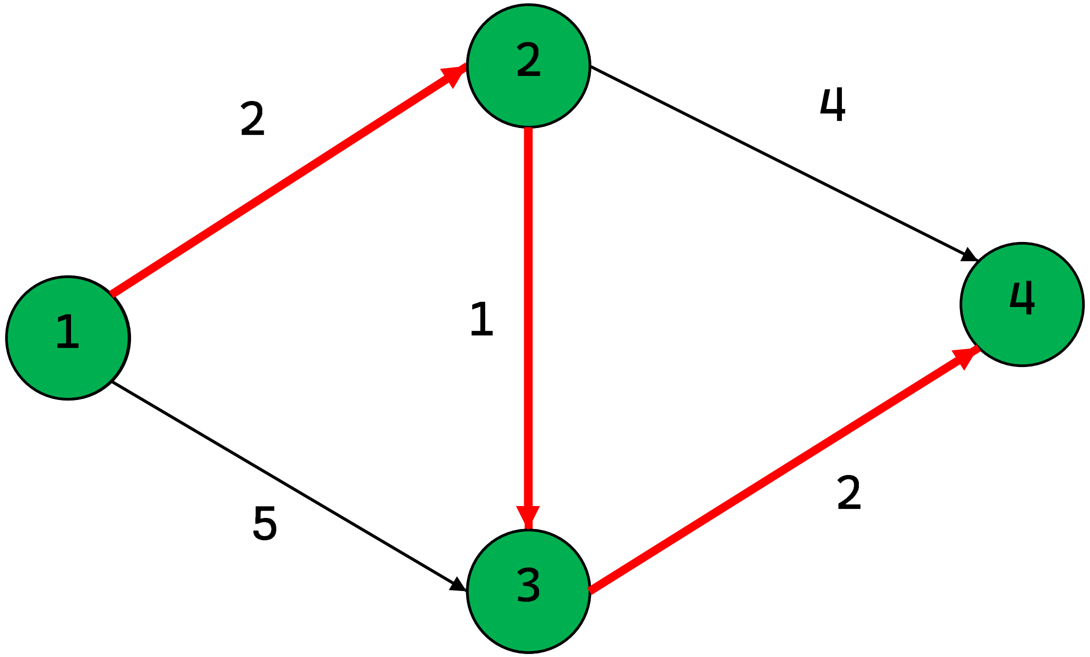



```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

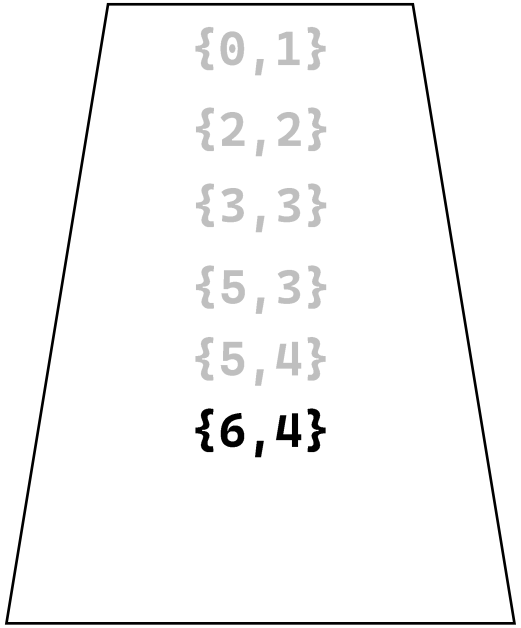
取出最小的：**{5,4}**，但 **4** 沒有連接到節點，所以沒做事

dis

index	1	2	3	4
value	0	2	3	5



priority_queue
{與 1 的距離,點}

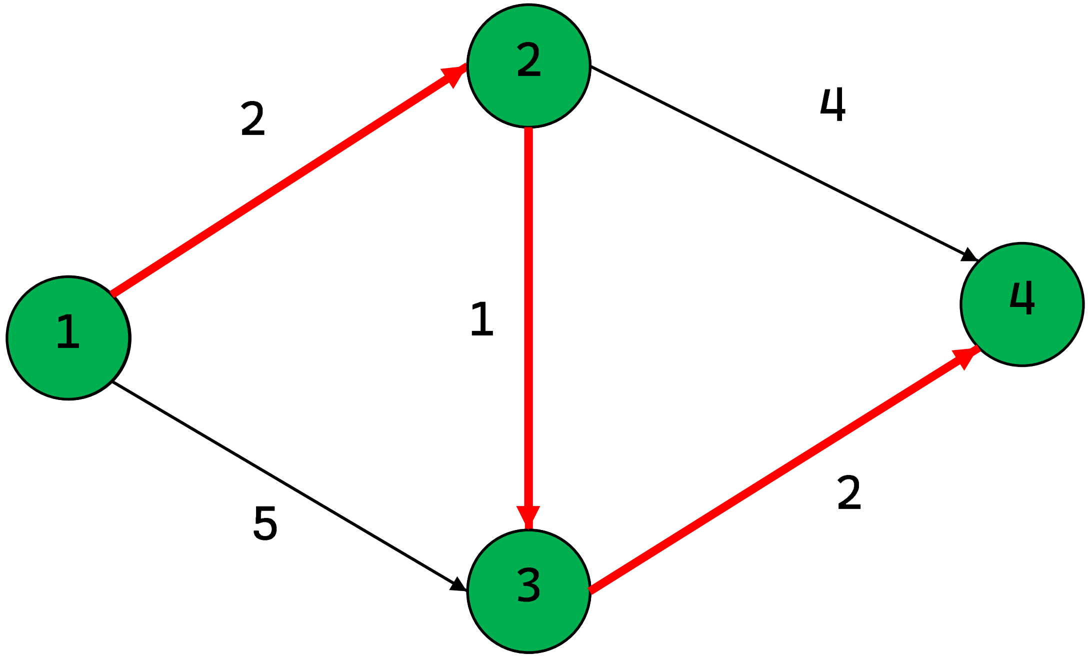


push {0,1}
pop {0,1}
dist[2] = 2 push {2,2}
dist[3] = 5 push {5,3}
pop {2,2}
dist[3] = 3 push {3,3}
dist[4] = 6 push {6,4}
pop {3,3}
dist[4] = 5 push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

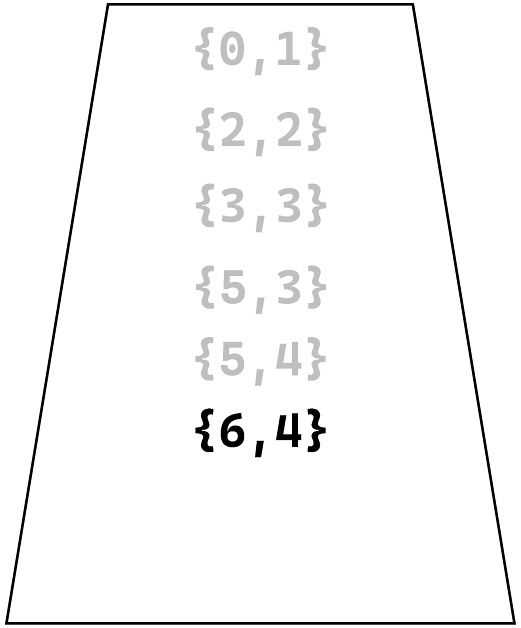
dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5

取出最小的：**{6,4} ，6 ≠ (dis[4] = 5) → continue** dis

index	1	2	3	4
value	0	2	3	5



priority_queue
{與 1 的距離,點}




```
push {0,1}
pop {0,1}
dist[2] = 2  push {2,2}
dist[3] = 5  push {5,3}
pop {2,2}
dist[3] = 3  push {3,3}
dist[4] = 6  push {6,4}
pop {3,3}
dist[4] = 5  push {5,4}
pop {5,3}
continue {5,3}
pop {5,4}
pop {6,4}
continue {6,4}

dis[1] = 0
dis[2] = 2
dis[3] = 3
dis[4] = 5
```

取出最小的：**{6,4}** ，**6 ≠ (dis[4] = 5) → continue**

完成